The **Mobile App** Project Canvas (ver. 1.0)

Connecting the Last Mile — Internet for All Territorians

CONCEPT

A hybrid solution addressing the Northern Territory's digital divide by:

a. *Developing an offline-first learning app to give remote students access to educational resources without continuous internet.*
b. *Deploying community-based mesh networks to provide affordable, reliable internet coverage in underserved areas.*

OBJECTIVES & PURPOSES

- *Reduce the digital access gap in pilot communities by at least 30% within the first year.*
- *Improve student access to curriculum content and telehealth services.*
- *Create a scalable model for other rural/remote regions.*

VALUES

1. *Accessibility – Inclusive design for all users, regardless of connectivity.*
2. *Affordability – Shared infrastructure lowers costs for individuals.*
3. *Community Empowerment – Locally managed and maintained networks*

PERSONAS

- **Remote Students** – Children and young adults in rural and Indigenous communities without stable internet access.
- **Teachers & Educators** – Upload and distribute learning materials that students can use offline.
- **Healthcare Workers** – Deliver telehealth services with low data usage.
- **Community Leaders & Councils** – Oversee deployment and sustainability of mesh networks.
- **Local IT Volunteers** – Assist in setup, troubleshooting, and maintenance.

PROBLEMS TO SOLVE

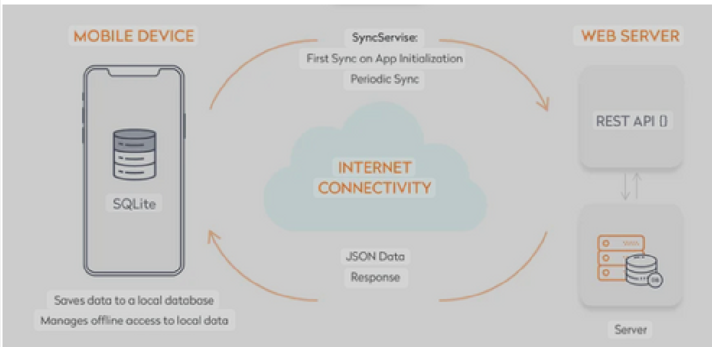
- **Inconsistent or non-existent internet in remote NT.**
- **High data costs making internet unaffordable.**
- **Education and telehealth access gaps for remote residents.**

COMPONENTS

1. **Offline Learning App –** Educational content, quizzes, video playback, and auto-sync.
2. **Mesh Network Hardware –** Low-cost routers, Raspberry Pi nodes, and satellite uplink.
3. **Management Dashboard –** For network monitoring, bandwidth allocation, and troubleshooting.

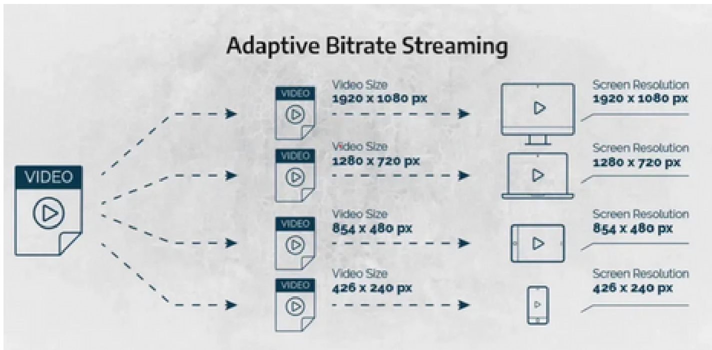
FEATURES

- *Works fully offline with sync-on-connect capability.*



- *Automatic mesh routing for uninterrupted local connectivity.*

- *Video compression and adaptive streaming to save bandwidth.*



- *Shared bandwidth among households to reduce costs.*
- *Easy network setup with plug-and-play nodes.*

MILESTONES

1. *Research & Planning – (2 weeks).*
2. *Prototype Learning App – (4 weeks).*
3. *Simulate Mesh Network – (2 weeks).*
4. *Pilot Deployment – (6 weeks).*
5. *Refinement & Expansion Plan – Incorporate feedback.*

STAKEHOLDERS

- **NT Government Digital Strategy & Education Departments**
- **Community Councils**
- **Schools & Training Centres**
- **Healthcare Services**
- **Technology Partners**

RISKS

- **Difficulty sourcing affordable, durable hardware.**
- **Limited local technical expertise for maintenance.**
- **Weather and environmental challenges for outdoor equipment.**
- **Funding sustainability for long-term operations.**

DELIVERABLES

- Progressive Web App (PWA) or APK for learning platform.
- Mesh network simulation and hardware setup documentation.
- Poster and presentation for CDU IT Code Fair 2025.
- Community training materials.

APP NAME

LAST MILE NT

CONTEXT OF USAGE &COVERAGE

- *Offline-first*
- *low-bandwidth, high-latency*
- single satellite or cellular uplink

TECHNOLOGY

- *Frontend: Flutter or React for PWA.*
- *Backend: Node.js with offline sync API.*
- *Network Simulation: NS-3 or Cisco Packet Tracer.*

PLATFORM,OS, ...

- *Android smartphones (primary).*
- *Web browsers (secondary).*

ORIENTATION

Both portrait and landscape support

RELEASE

- *local council partnership.*
- *Public r free app with open-source mesh network guidelines.*